

VOLCARCH — A Short Brief

The missing-archaeology problem behind the VOC-NLP proposal

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Prepared as a quick background read for Prof. Suzan Verberne, LIACS, Leiden University

ONE-PARAGRAPH SUMMARY

VOLCARCH is a multi-paper project I have been developing over the last ~18 months that quantifies, for the first time with calibrated Bayesian uncertainty, how much of the pre-colonial archaeological record in volcanic Indonesia is physically unrecoverable by conventional survey — and therefore how much of pre-1600 Javanese, Sumatran, and Balinese settlement history has to be reconstructed from other evidence streams. It is the archaeological motivation behind the VOC-NLP PhD proposal: the reason the colonial text archive matters is that the dirt archive is, for a quantifiable majority of sites, gone.

THE CORE RESULT

Across 51 calibrated eruption-site pairs drawn from three volcanic systems (Merapi-Merbabu, Agung-Batur, Tambora-Rinjani), the mean tephra sedimentation rate over pre-colonial Indonesia is 4.4 ± 1.2 mm/yr ($r = 0.951$). Projected over a 1,000-year pre-colonial window, the expected burial depth at typical settlement locations is ~7 metres. Under conventional archaeological survey methods (surface survey, shallow test pits, coring at ≤ 2 m), this implies that ~99.94% of expected pre-1600 settlement signal in the study region is archaeologically invisible — not destroyed, but buried below standard detection thresholds.

This is not a headline-chasing number. It is the output of 175 tracked experimental configurations, 100,000 Monte Carlo runs for robustness, and an FDR audit in which 65 of 83 tests survived Benjamini–Hochberg correction (78.3%). The full experimental portfolio and data are on Zenodo (DOI: 10.5281/zenodo.19081502).

WHY THIS MATTERS FOR NLP

If 99.94% of the expected settlement record is physically inaccessible, then any reconstruction of pre-colonial Indonesian settlement patterns has to come from textual, epigraphic, or remote-sensing sources. The textual source with by far the highest volume and density is the VOC archive: ~22,000 administrative references to Indonesian settlements between 1602 and 1800, most of them never converted into structured data.

The VOC-NLP PhD proposal is the natural follow-up: it asks whether NLP can convert that archive into the structured settlement record that archaeology cannot recover, and — critically — whether the VOLCARCH sediment model can act as an independent physical validation signal for textual extractions. Extracted settlements in areas the model predicts should be buried are either (a) IE errors or (b) genuine historical surprises. Both are interesting.

This is, to my knowledge, an unusual position for historical NLP: a concrete, physically-grounded, non-textual ground truth.

CURRENT PUBLICATION STATUS

Four papers from the VOLCARCH series are currently under review at Q1 venues:

- EGQSJ (E&G Quaternary Science Journal) — the core taphonomic framework and cascade model
- JCAA (Journal of Computer Applications in Archaeology) — GIS-based settlement suitability modelling
- Antiquity (Project Gallery) — Temporal Overlay Matrix for archaeological visibility
- Oceanic Linguistics — lexical-change companion (arXiv:2604.00023)

Three more are in drafting:

- Archipel (INALCO/EHESS) — settlement hierarchy gaps in 9th–10th century Java
- Digital Humanities Quarterly — computational textual archaeology
- Archeologia e Calcolatori — sacred vs. administrative space in the epigraphic record

These papers are background that motivates the PhD and establishes I can carry a multi-paper research programme from design through peer review. The precise role they play in a cumulative dissertation (background chapters vs. self-contained thesis chapters) is something to agree with the host institution.

OTHER RESEARCH RELEVANT TO AN NLP PhD

Beyond VOLCARCH, my NLP portfolio includes:

- KorupsiNLP — a corpus of 693 Indonesian Supreme Court corruption verdicts, sentencing prediction work, 50+ autoresearch experiments. Two papers submission-ready.
- NusantaraAgent — neuro-symbolic legal reasoning (ASP + LLM) for Indonesian customary law; 95 expert-verified rules; ASP+DeepSeek 68.6% accuracy.
- ModernKataKupas — a published Indonesian morphology library (PyPI v1.0.1, 29,936 root words, 93 tests). Open-source, in use.

CODE AND DATA AVAILABILITY

Everything is public. Links for reference:

- VOLCARCH — code + configs: github.com/neimasilk/volcarch-repo | data: Zenodo DOI 10.5281/zenodo.19081502
- KorupsiNLP — github.com/neimasilk/korupsiNLP
- NusantaraAgent — github.com/neimasilk/nusantara-agent
- ModernKataKupas — github.com/neimasilk/modern_kata_kupas (PyPI: `pip install modernkatakupas`)

A NOTE ON METHODOLOGY AND AI

I want to say this explicitly: I do not read 17th-century Dutch, and the VOC-NLP proposal is designed around that fact. The methodology treats modern NLP tooling as a primary instrument, with historian and linguist collaboration for validation. I am a computational researcher, not a historical philologist, and the honest framing of this kind of heritage-NLP PhD is exactly that — a collaboration between NLP engineering and domain expertise, structured around auditable outputs. The fuller PhD proposal will make this stance explicit.

WHAT COMES NEXT

A proper PhD proposal — scoped to LIACS, aligned with EXALT and “Digging in Documents”, with RQ structure, thesis outline, timeline, and funding plan — will follow within one to two working days. This brief exists so that there is something concrete to read first, rather than a proposal assembled in haste.

Thank you for your time and your thoughtful reply.

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